PROJECT OVERVIEW

Quality of Life Per Dollar

Which Cities Let You Live Well Without Breaking the Bank?

Housing & Cost

Analyzing affordability ratios and the true cost of residential comfort.

Health Outcomes

Evaluating healthcare quality and public health metrics across global regions.

Transport & Mobility

Measuring traffic congestion and infrastructure efficiency for urban living.

Data Journalism | Group 6 | Daniel, Wilson, Arhan, Colin

The Liveability Paradox

Every year, the Economist Intelligence Unit (EIU) ranks 173 cities on stability, healthcare, culture, education and infrastructure. Copenhagen topped the 2025 index; Damascus ranked last. Cities such as Vienna, Zurich, Melbourne, and Geneva dominate the top 10, but more importantly, all 10 of the most liveable cities also rank among the world's most expensive.

Existing indices measure what a city offers, but not what it what it costs to actually enjoy them. Our project fills that gap with an original "value for living" index.

OUR APPROACH

Combine Numbeo, WHO & World Bank data to rank cities by quality of life per dollar spent.

173
Cities ranked globally

#1
Copenhagen 2025

5
Categories measured

EIU Index Methodology

Stability — 25%
Crime · Terrorism · Civil disorder · War threat

Healthcare — 20%
Public & private care · Drug access · WHO indicators

Culture & Environment — 25%
Climate · Corruption · Censorship · Food & sport

Education — 10%
Private schooling quality · World Bank state indicators

Infrastructure — 20%
Transport · Housing · Energy · Water · Telecoms

What We're Asking

01

Which cities offer the highest quality of life relative to cost of living?

Building an original 'value for living' index across 500+ global cities

02

Are ‘good value’ countries clustered in particular world regions and what are indicators of such countries?

Creating a country level value for living index and grouping by different world regions.

03

What is the relationship between city level traffic and pollution?

Decomposition analysis across our composite index dimensions

04

Does income inequality explain why some low-cost countries still score poorly on quality of life?

Cross-referencing World Bank Gini coefficient against our value for living index

05

Has the cost of living risen faster than quality of life since COVID? (Country)

Longitudinal trend analysis using Numbeo historical data from 2012 to present

Our Data Sources

1

Numbeo Datasets, Country and City Level

Core datasets

Key datasets: Cost of Living Index, Crime Index, Health Care Index, Pollution Index, Property Price Index, Traffic Index

2

WHO Global Health Observatory

Health quality

Key variables: Life expectancy at birth · Healthy life expectancy at birth (HALE) · By country, 2000–present

3

World Bank: World Development Indicators

Economic context layer

Key variables: GDP per capita PPP · Gini index · Life expectancy · Health expenditure per capita · 217 countries

Analysis Conducted

01

Building the Index

- Combine Numbeo sub-indices (health, safety, traffic, pollution) into a composite quality score at both city and country level
- Divide quality score by cost of living to produce a 'value for living' ratio per city
- Apply a quality floor so rankings only include cities above the 25th percentile quality score
- Bar chart of top 30 cities and heatmap of dimension scores

02

Geographic Patterns

- Choropleth world map coloured by value for living index
- Regional breakdown: which parts of the world have high liveability, low cost cities?
- Scatter plot of value index vs. GDP per capita: finding the middle-income sweet spot
- Value quadrant: every country plotted by quality vs cost, identifying sweet spots, prestigious, budget, and trap countries

03

Dimension Decomposition

- Comparing WHO life expectancy vs. Numbeo healthcare index to identify where perception diverges from outcomes
- Rent burden analysis: rent as a share of local purchasing power
- Analysis of the relationship between city-level traffic and pollution across regions
- Income inequality analysis: analysing why some low-cost countries still score poorly on liveability

04

City Profiles & Time Series

- Affordability ratio trends from 2016 to 2024 tracking whether residents kept pace with rising costs
- COVID impact analysis: four-act narrative covering the pre-COVID baseline, the 2020-2021 shock, the recovery, and the net verdict by 2024
- League table of cities with the largest affordability gains and declines since COVID

Key Insights Found

01

Research Question 1

- Top-value cities concentrated in Eastern Europe (Russia, Ukraine) and South Asia (India, Pakistan), with quality scores above 52 combined with cost of living indices well below 25 on the NYC-anchored scale
- EIU top liveability cities (Vienna, Copenhagen, Zurich) rank in the bottom 10% of our value index: Vienna 106th, Zurich 217th of 222 cities, as quality is high but costs are prohibitive
- Top-value cities are not just cheap: they score above average on safety (+5.2) and low traffic (+7.6), though healthcare is average, suggesting safety is the key quality differentiator.

02

Research Question 2

- South Asia and Eastern Europe are the highest-value regions:
  - South Asia leads on cost (avg CoL 23),
  - Eastern Europe leads on quality (avg quality score 64)
- GDP per capita is strongly negatively correlated with value index ($r = -0.56$), confirming that higher income countries are systematically more expensive relative to their quality and validating the middle-income sweet spot hypothesis
- EIU top countries rank at the very bottom of our index: Switzerland ranks last (89th of 89) with a value index of 0.77 compared to Russia's 2.52, despite Switzerland having higher quality scores.

Key Insights Continued

03

Research Question 3

- **Traffic and pollution are moderately but not strongly correlated (r ≈ 0.43–0.58)** — they are related but distinct dimensions, justifying keeping both in the quality index rather than collapsing them into one
- **Commute time is the strongest predictor of pollution** (r = 0.482) — stronger than even the CO2 index, suggesting overall urban congestion stress matters more than raw vehicle emissions
- **About 1 in 5 cities decouple high traffic from high pollution** — mostly North American and Gulf cities with stricter emission standards or geographic advantages

04

Research Question 4

- Inequality matters globally. High-Gini countries score below what their cost of living predicts (r = -0.350, p = 0.002).
- But among low-cost countries specifically, inequality alone explains nothing (p = 0.52).
- What separates poor performers is safety (13.6 points lower) and traffic congestion (23.4 points lower), not healthcare or air quality.

05

Research Question 5

- Affordability declined sharply during the COVID shock, with the median affordability ratio falling 28% from 1.56 in 2019 to 1.12 in 2021 as purchasing power collapsed faster than costs adjusted
- By 2024, affordability had not recovered: 101 of 125 cities remain less affordable than in 2019, with every world region showing a negative median change
- Quality of life scores were largely resilient throughout, finishing within 2.4 points of 2019 levels across all four dimensions, confirming this is a cost and wage story rather than a deterioration in services

Stories Emerging from the Analysis

1

Good Value Cities are Not What we Expect

Eastern Europe and South Asia dominate the top of the value index at both city and country level. These places are not just cheap. They clear a genuine quality floor on safety and infrastructure. The cities and countries that score highest are not the ones you would find in any existing liveability ranking, because every existing ranking ignores what things actually cost for local residents.

2

Wealth buys quality, but at a rapidly diminishing return

The GDP vs value index correlation (r = -0.56) shows that as countries get richer, costs rise much faster than quality improves. Switzerland, Denmark, and Australia have high quality scores, but they rank at the very bottom of our value index. The EIU and Mercer rankings are essentially measuring wealth, not value, and the two diverge sharply once you account for what residents actually pay.

3

COVID Eroded Affordability and it has not Recovered

The affordability ratio fell 28% between 2019 and 2021 and remained below pre-COVID levels for 101 of 125 cities by 2024. Crucially, quality of life scores barely moved across all four dimensions. Safety, healthcare, clean air, and traffic all finished within 2.4 points of 2019. The post-COVID cost of living crisis is a wages and prices story, not a services story. People are not living worse, but they are just paying more relative to what they earn.

Next Planned Steps

1

Step 1 - Validate Findings Against Country Case Studies

Compare our quantitative results with short qualitative case studies of countries that emerged as outliers, such as Costa Rica, Lebanon, and Nigeria. This will test whether our statistical predictions align with on-the-ground reporting and lived experience, strengthening confidence in the value index and exposing where perception-based data may mislead.

2

Step 2 - Consolidate all Research Question Notebooks into one Complete Analysis

Synthesize findings from RQ1 through RQ5 into a unified value-for-living framework, anchored by a single executive summary. This will tie together the redundancy tests, the inequality residuals, the post-COVID trajectory clusters, and the regional breakdowns into one cohesive narrative ready for presentation and final reporting.

3

Step 3 - Create an Interactive Website to Visualize Key Analyses

Build a public-facing website that lets users explore the dataset on their own terms. Visitors can filter by region, year, or price range, walk through our key visualizations, and replicate the cost vs quality and inequality residual analyses interactively, making the project accessible beyond the static notebook format.